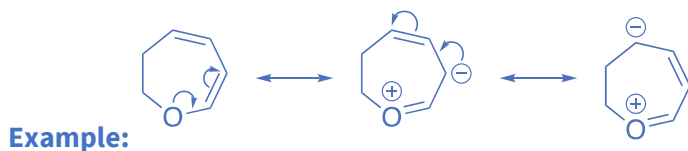
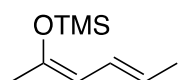


Delocalized Pi Systems and Aromatic Compounds

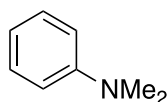
1. Each of the following molecules is nucleophilic at multiple atoms. Provide the indicated number of alternative resonance forms for the molecule to reveal the nucleophilic atoms. Illustrate electron movements between the resonance forms with curved arrows.



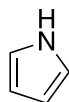
- A. Two resonance forms



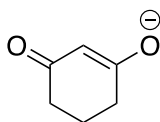
- B. Three resonance forms



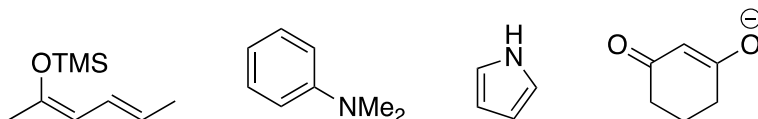
- C. Four resonance forms



- D. Two resonance forms

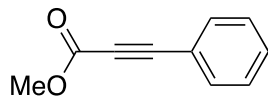


2. Build the delocalized pi systems of each of the molecules above in [HuLiS](#) and use the pi molecular orbitals to determine the *most* nucleophilic atom in each molecule. Circle this atom on each structure.

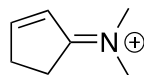


3. Each of the following molecules is electrophilic at multiple atoms. Provide the indicated number of alternative resonance forms for the molecule to reveal the electrophilic atoms.

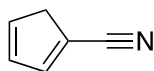
- A. Four resonance forms



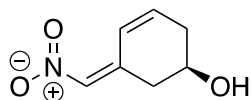
- B. Two resonance forms



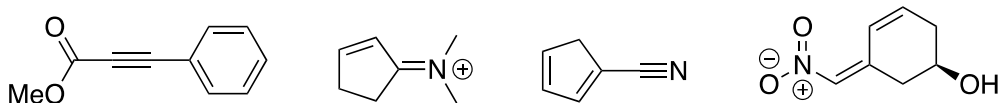
- C. Three resonance forms



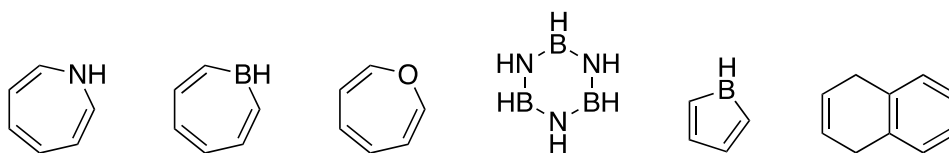
- D. Two resonance forms



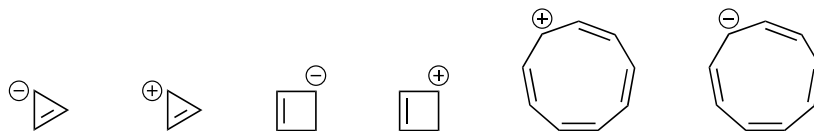
4. Build the delocalized pi systems of each of the molecules above in [HuLiS](#) and use the pi molecular orbitals to determine the *most* electrophilic atom in each molecule. Circle this atom on each structure.



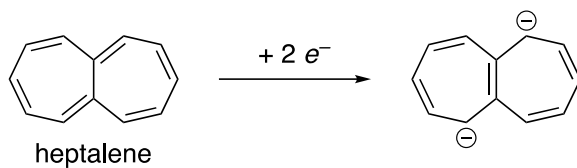
5. Provide structures for hypothetical compounds that fit the following descriptions.
- A. A disubstituted benzene containing two electron-donating groups in an *ortho* relationship
 - B. A disubstituted benzene containing an electron-donating group and electron-withdrawing group in a *para* relationship
 - C. A trisubstituted benzene containing three electron-donating groups at the 1-, 3-, and 5-positions of the benzene ring
6. Indicate which of the following compounds is aromatic, assuming planarity for the necessary set of atoms in each structure.



7. Indicate which of the following ions is aromatic, assuming planarity for the necessary set of atoms in each structure.

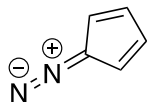


8. Heptalene is a nonplanar molecule but becomes planar upon two-electron reduction, which yields a dianion.



- Identify *planar* heptalene as aromatic or antiaromatic and provide a justification for your conclusion.
- Identify the planar heptalene dianion as aromatic or antiaromatic and provide a justification for your conclusion.
- Using your answers to the previous two problems, explain why reduction causes planarization of heptalene.

9. Most diazo compounds are extremely unstable, to the point of being explosive. Diazocyclopentadiene is a notable exception. Provide a significant alternative resonance form of the structure below and use it to justify the stability of this compound.



diazocyclopentadiene

10. Pyridine and imidazole are mild bases at their N2 nitrogen atoms—the pK_a of pyridinium is 5.25 and that of imidazolium is 7.38. Pyrrole is not at all basic and protonation occurs at *carbon* rather than nitrogen.



pyridine



imidazole

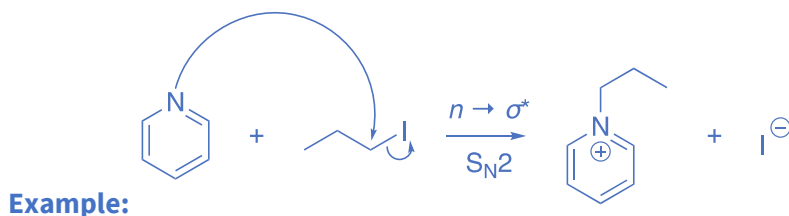


pyrrole

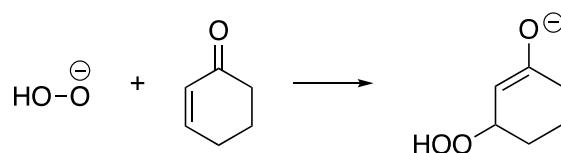
- Provide structures for the conjugate acids of these three compounds. Carefully consider which carbon in pyrrole is protonated!
- Explain why pyridine and imidazole are much more basic than pyrrole.
- Explain why protonation occurs at carbon rather than nitrogen in pyrrole.

Mechanisms of Organic Reactions

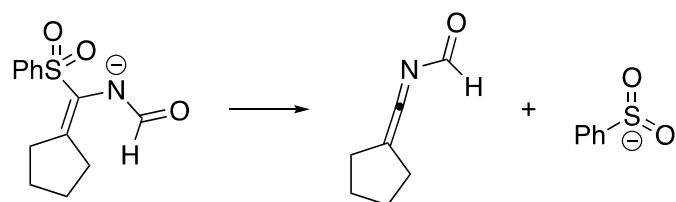
11. Each reaction shown below occurs in a single elementary step. For each step, draw curved arrows that convert the starting structure(s) exactly into the product structure(s). Use donor and acceptor orbitals to create an orbital interaction " $x \rightarrow y$ " and label the elementary step.



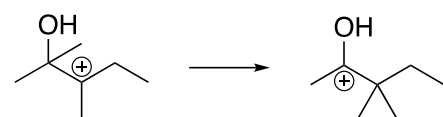
A.



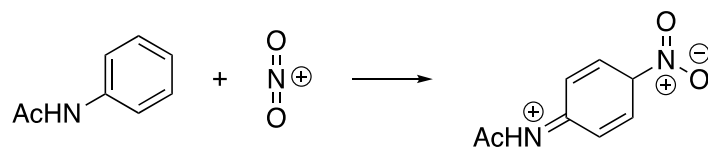
B.



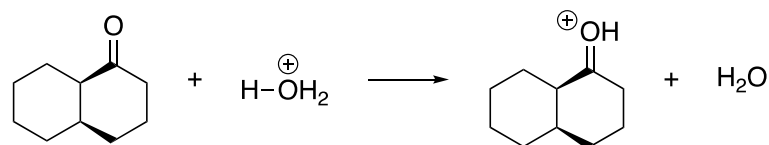
C.



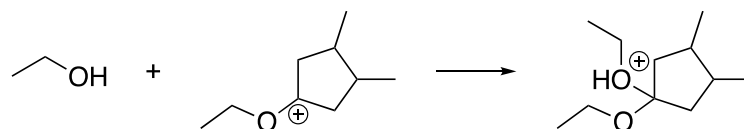
D.



E.

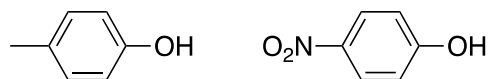


F.

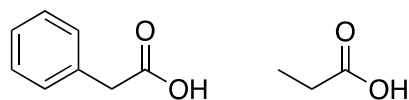


12. For each elementary step above, draw the transition state. Use dotted lines to represent partial bonds and indicate the partial charges of atoms whose charges change as the step proceeds using δ^+ and δ^- .
13. For each pair of compounds below, determine the more acidic compound and provide a rationalization of the difference in acidity between the two compounds. Appeal to a key structural stability factor in each case.

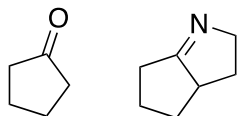
A.



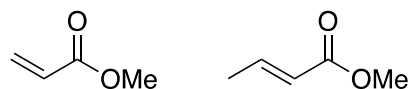
B.



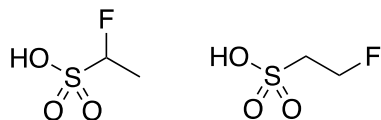
C.



D.

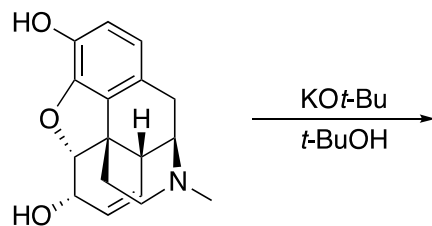


E.

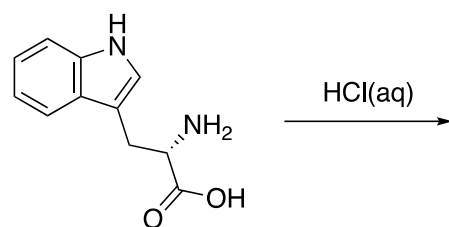


14. Each reaction below includes a strongly acidic or basic reagent. Draw the products of proton transfer to or from the organic substrate in each case.

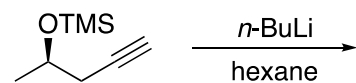
A.



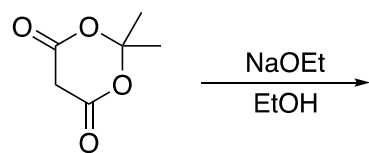
B.



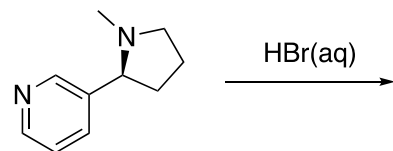
C.



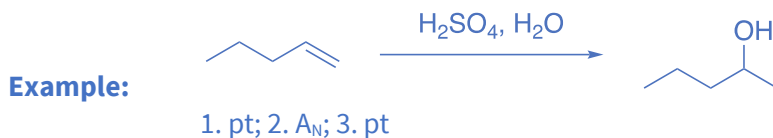
D.



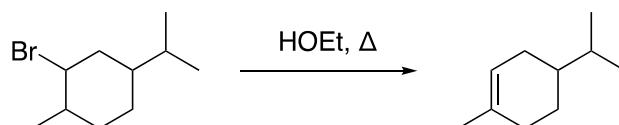
E.



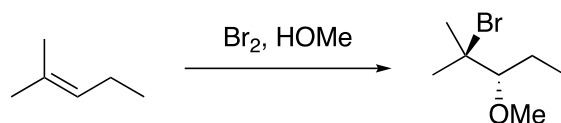
15. The reactions below are drawn from classes of reactions you've encountered in Organic Chemistry I. Indicate the mechanism of each reaction using a series of elementary step labels. While you're welcome to draw curved arrows when thinking through this, do not include a curved-arrow mechanism with your response.



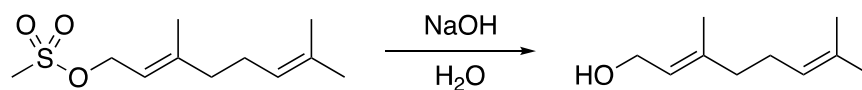
A.



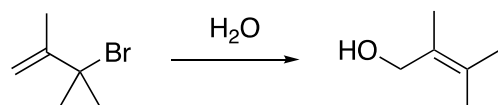
B.



C.



D.



E.

